

Consumption Sorting and Inequality

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Motivation

The top 10% earners now account for 50% of total personal spending (Moody's, 2025).

- Shopping behavior matters. For the same goods:
 - [Kaplan and Menzio \(2015\)](#): shopping behavior affects prices paid;
 - [Sangani \(2022\)](#), [Nord \(2022\)](#): high earners spend less effort shopping, pay more;
- Similar households cluster together:
 - [Gaubert et al. \(2021\)](#): segregation of high earners $\Rightarrow$ regional income divergence;
 - [Boar and Giannone \(2025\)](#): large consumption segregation, correlated with income's;

$\Rightarrow$ What is the interplay of spatial segregation and shopping externalities?

$\Rightarrow$ How do these forces shape the distributional impacts of rising prices?

Today

We develop and quantify a dynamic spatial model with search in the product market:

- Heterogeneous households with non-homothetic preferences choose where to live;
- Retailers also choose location, targeting local demand;
- Frictional market for non-durables: retailers post prices, households exert search effort.

We find evidence of the model mechanisms in the data.

We calibrate the model to match evidence on spatial segregation and price dispersion.

We leverage the calibrated model to study a uniform increase in non-durable marginal costs.

Environment

Household problem is divided in two stages:

- ① Households solve a consumption/saving problem, rent housing, choose location;
- ② Given the allocated consumption budget from the first stage,
 - purchase homogeneous good c_0 ;
 - purchase J non-tradeable heterogeneous goods (groceries);
 - decide how much effort to put in search.

Retailers:

- Choose location, pay flow rental cost $K_{\ell,j}$ to operate;
- Sell one variety, maximize profits posting prices $\sim F_j^\ell(p)$.

Markets:

- Perfectly competitive market for c_0 , frictional market for each grocery variety j ;
- Absentee landlords supply residential and commercial housing in separate markets;
- Perfectly competitive labor market, linear production in labor.

Household problem

First stage: households choose expenditure e , housing h , savings s , and location ℓ .

$$V(z, a, \ell) = \max_{e, h, a'} U(e, \ell) + \tilde{u}(h) + A_\ell + \beta W(z', a', \ell),$$

$$W(z, a, \ell) = \max_{\ell' \in L} \left[\mathbb{E}_z V(z', a, \ell') + \gamma^{\ell'} \right],$$

s.t.

$$e + q_\ell h + a' = a(1 + r) + z w,$$

$$z' = \rho_z z + \sigma_z \varepsilon^z$$

$$\gamma^\ell \sim \text{Gumbel}(\lambda).$$

Household problem - Optimality conditions

Euler:

$$U_e(e, \ell) = \beta(1 + r)\mathbb{E} U_e(e', \ell').$$

Housing choice (static):

$$U_e(e, \ell) = \tilde{u}_h(h)/q_\ell.$$

Location shares:

$$\pi_j = \frac{\exp(V(z, a, j)/\lambda)}{\sum_k \exp(V(a, z, k)/\lambda)}.$$

Shopping problem

Second stage: household chooses search effort s and purchase non-durables c .

$$U(e, \ell) = \max_{s, \{c_j\}_{j=1}^J, c_0} \frac{\mathcal{C}^{1-\phi}}{1-\phi} - v(s, \mathcal{C}),$$

s.t.

$$\mathcal{C} = (c_G)^\alpha (c_0)^{1-\alpha}$$

$$c_G = \left(\sum_{j=1}^J \mathcal{C}^{q_j/\sigma} c_j^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}$$

$$c_0 + \sum_{j=1}^J p_j(s, \ell) c_j \leq e$$

Shopping problem - optimality conditions

Optimal search effort:

$$-v_s(s, \mathcal{C}) = \underbrace{\frac{\partial \mathbb{P}(\mathcal{C}, s, \ell)}{\partial s} \mathcal{C}}_{\text{change in BC}} \underbrace{\frac{\mathcal{C}^{-\phi} - v_c(s, \mathcal{C})}{\mathbb{P}(\mathcal{C}, s, \ell) + \frac{\partial \mathbb{P}(\mathcal{C}, s, \ell)}{\partial \mathcal{C}} \mathcal{C}}}_{\text{MB of relaxing BC}}$$

Optimal consumption by variety:

$$c_j = \alpha e \mathcal{C}^{q_j} p_j(s, \ell)^{-\sigma} p_G(\mathcal{C}, s, \ell)^{\sigma-1}$$

where

$$p_G(\mathcal{C}, s, \ell) = \left[\sum_j \mathcal{C}^{q_j} p_j(s, \ell)^{1-\sigma} \right]^{1/(1-\sigma)}$$

$$\mathbb{P}(\mathcal{C}, s, \ell) = p_G(\mathcal{C}, s, \ell)^\alpha \alpha^{-\alpha} (1 - \alpha)^{\alpha-1}$$

Search problem

Each c_j is the CES aggregation of countless transactions; household solve

$$\begin{aligned} \min_{\{c_{j\tau}\}} \int_0^1 p_{j\tau} c_{j\tau} d\tau \\ \text{s.t} \\ c_j = \left(\int_0^1 (c_{j\tau})^{(\sigma_j-1)/\sigma_j} d\tau \right)^{\sigma_j/(\sigma_j-1)} \end{aligned}$$

For each transaction, the **household can meet either one or two retailers**.
Probability of meeting two retailers given by matching function $m_j(s, \ell)$.

Retailer problem - Posting

A retailer operating in location ℓ and producing good j maximizes

$$\pi_j(p, \ell) = (p - \kappa_j) \int \lambda(e, \ell) \frac{(1 - m_j(s(e), \ell)) + m_j(s(e), \ell) 2(1 - F_j^\ell(p))}{M_\ell^j} \times \\ \times \left[\left(\frac{p}{p_j(s(e), \ell)} \right)^{-\sigma_j} c_j(e, \ell) \right] de - K_{\ell, j},$$

where the price index for good j is

$$p_j(s, \ell) = \left\{ [1 - m_j(s, \ell)] \int p_{j\tau}^{1-\sigma_j} dF_{\ell, j}(p_{j\tau}) + m_j(s, \ell) \int p_{j\tau}^{1-\sigma_j} dF_{\ell, j}^m(p_{j\tau}) \right\}^{\frac{1}{1-\sigma_j}}.$$

Retailer problem

Aggregate meeting rate definition

A no-mass distribution F with continuous support emerges:

$$F_{\ell,j}^{\ell}(p) = 1 - \frac{1 - \bar{m}_{\ell,j}}{2\bar{m}_{\ell,j}} \left[\left(\frac{\bar{p}_j}{p} \right)^{-\sigma_j} \frac{\bar{p}_j - \kappa_j}{p - \kappa_j} - 1 \right] \quad \forall p \in [\underline{p}_{\ell,j}, \bar{p}_j]$$

where $\bar{p}_j = \kappa_j \frac{\sigma_j}{\sigma_j - 1}$ and $\underline{p}_{\ell,j}$ is endogenous.

Retailer problem

Free entry drives ex-ante profits to zero. Number of retailers is determined as:

$$M_{\ell,j} = \frac{\bar{c}_{\ell,j}}{K_{\ell,j}} (1 - \bar{m}_{\ell,j}) \bar{p}_j^{-\sigma_j} (\bar{p}_j - \kappa_j),$$

Retailers need commercial housing to operate, they rent floor space at unit rent q_ℓ^C :

$$K_{\ell,j} = q_\ell^C \underline{h}.$$

Housing and labor markets

Absentee landlords supply housing according to these inverse supply functions:

$$\begin{aligned}q_{\ell} &= \eta_0^{\ell} (H_{\ell}^S)^{\eta_1^{\ell}}, \\ q_{\ell}^C &= \eta_C (H_{\ell}^{C,S})^{\eta^C}.\end{aligned}$$

- Firms sell groceries to retailers at competitive prices;
- Firms operate linear production functions, $y_O = Z_O$ and $y_j = Z_j/\kappa_j$;
- Firms do not congest the housing market.

Empirical agenda

- ① Relationship between search intensity and prices paid (for the same products),

$$p_j(s, \ell) = \left\{ [1 - m_j(s, \ell)] \int p_{j\tau}^{1-\sigma_j} dF_j^\ell(p_{j\tau}) + m_j(s, \ell) \int p_{j\tau}^{1-\sigma_j} dF_j^m(p_{j\tau}) \right\}^{\frac{1}{1-\sigma_j}};$$

- ② Relationship between income and search effort,

$$-v_s(s, \mathcal{C}) = \frac{\partial \mathbb{P}(\mathcal{C}, s, \ell)}{\partial s} \mathcal{C} \frac{\mathcal{C}^{-\phi} - v_c(s, \mathcal{C})}{\mathbb{P}(\mathcal{C}, s, \ell) + \frac{\partial \mathbb{P}(\mathcal{C}, s, \ell)}{\partial \mathcal{C}} \mathcal{C}};$$

- ③ Relationship between location characteristics and price paid (for the same products),

$$F_{\ell,j}^\ell(p) = 1 - \frac{1 - \bar{m}_{\ell,j}}{2\bar{m}_{\ell,j}} \left[\left(\frac{\bar{p}_j}{p} \right)^{-\sigma_j} \frac{\bar{p}_j - \kappa_j}{p - \kappa_j} - 1 \right];$$

- ④ Relationship between location characteristics and search intensity.

Data description

Kilts-NielsenIQ Consumer Panel dataset:

- tracks 40,000-60,000 American households;
- panelists use in-home scanners or mobile apps to register info about grocery purchases;
- every purchase of each product is linked to a specific shopping trip and barcode;
- includes socio-demographic characteristics of the panelists.

Other cross sectional datasets for county-level information:

- County Business Patterns (CBP): establishment counts;
- American Community Survey (ACS): demographics.

Price index definition

Products j (barcode) are bought by households i , on day t in period m .

$$\bar{p}_{j,m} = \sum_{i, t \in m} w_{i,m} p_{j,t}^i \left(\frac{q_{i,t}^j}{\bar{q}_m^j} \right), \quad (\text{average price})$$

$$Q_m^i = \sum_{j, t \in m} \bar{p}_{j,m} q_{i,t}^j, \quad (\text{counterfactual expenditure})$$

$$\bar{P}_{i,m} := \frac{\left(\sum_{j, t \in m} p_{i,t}^j q_{i,t}^j \right) / Q_m^j}{\sum_{i'} w_{i',m} \left(\sum_{j, t \in m} p_{i,t}^{j'} q_{i,t}^{j'} \right) / Q_m^{i'}}. \quad (\text{price index})$$

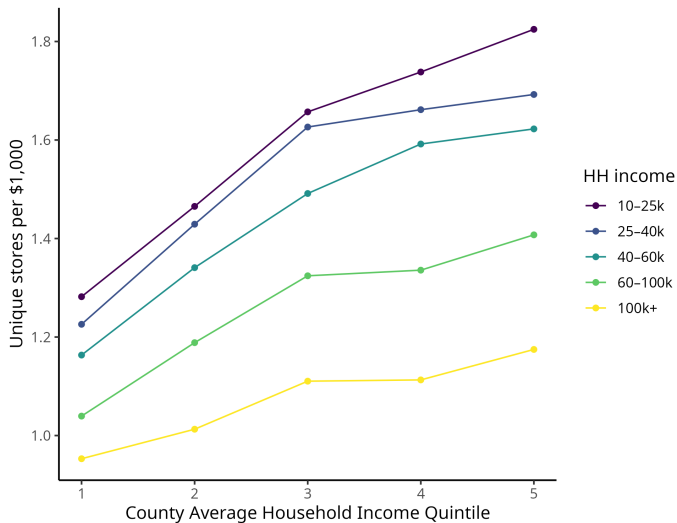
Great deal of variation! IQR 10pp, 90th-10th percentiles 20pp.

Price index and county average income

	Log(Price Index)		
Log(HH Income)	0.0201*** (0.0013)		0.0197*** (0.0013)
Log(County Income)		0.0175*** (0.0035)	0.0093** (0.0035)
R ²	0.03707	0.03107	0.03722
Observations	6,879,064	6,879,064	6,879,064

Standard-errors in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1
 All regressions control for Head Age and Sex, Employment status,
 Household composition, Month and Year

Shopping effort, income, location



Search intensity on store density

Per dollar

	Log(Unique Stores Visited)		Log(Trips)	
Log(Stores/km ²)	0.0417*** (0.0012)	0.0391*** (0.0013)	0.0208*** (0.0017)	0.0237*** (0.0018)
Log(Income)	-0.0299*** (0.0030)	-0.0338*** (0.0030)	-0.1721*** (0.0043)	-0.1676*** (0.0043)
Observations	6,535,161	6,535,161	6,535,161	6,535,161
R ²	0.16579	0.16662	0.28959	0.29013

Standard-errors in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

All regressions control for Head Age and Sex, Employment status,
Household composition, Expenditure, Month and Year

Prices on search intensity

Appendix

	Log(Price Index)			
Log(Unique Stores Visited)	-0.0257*** (0.0006)	-0.0259*** (0.0006)		
Log(Trips)			-0.0343*** (0.0006)	-0.0343*** (0.0006)
Log(County Income)		0.0122*** (0.0025)		0.0072*** (0.0024)
Observations	6,879,991	6,879,991	6,879,991	6,879,991
R ²	0.05355	0.05380	0.07000	0.07009

Standard-errors in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

All regressions control for Head Age and Sex, Employment status,
Household composition, Expenditure, Month and Year

Parametrization

- The flow utility from housing is CRRA, with parameter ϕ ;
- There are three grocery varieties, $J = 3$, and two locations;
- Locations match the below and above median counties in terms of average income;
- The matching function takes as inputs search effort and number of firms:

$$m_j(s, \ell) = s^\theta (B M_{\ell,j})^{1-\theta};$$

- The search cost function follows:

$$v(s, \mathcal{C}) = \psi_0 \mathcal{C}^{\psi_1} \frac{s^2}{1-s}.$$

We follow [Nord \(2022\)](#) for all externally calibrated parameters.

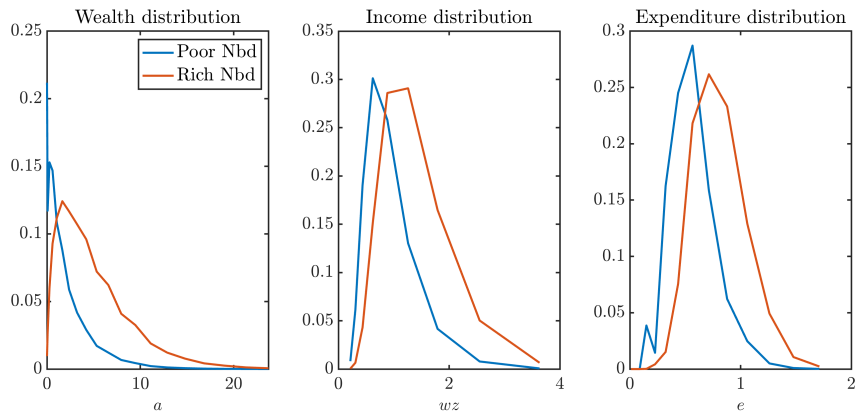
Calibrated Parameters

Parameter		Value
η_1^1	inverse elasticity of residential housing supply, Periphery	1.99
η_1^2	inverse elasticity of residential housing supply, Core	2.09
η_1^C	inverse elasticity of commercial housing supply	2.04
η_0^1	scale parameter for residential housing supply, Periphery	0.043
η_0^2	scale parameter for residential housing supply, Core	0.108
λ	Gumbel dispersion	0.095
α_H	Preference shifter for housing	0.95
$\underline{h}$	Floor requirement retailers	0.82
A_1, A_2	Amenities	0, 0.52
θ	Elasticity of matching function to search effort	0.92
β	Discount Factor	0.955
$\sigma_1, \sigma_2, \sigma_3$	Within Variety EOS	1.34, 1.37, 1.41
q_1, q_2, q_3	Non-homotheticity parameters	-1.66, -0.83, 0
$\kappa_1, \kappa_2, \kappa_3$	Marginal costs	1, 1.44, 1.88
ψ_1	Search cost scale parameter	0.011
ψ_2	Search cost elasticity to consumption	0.764

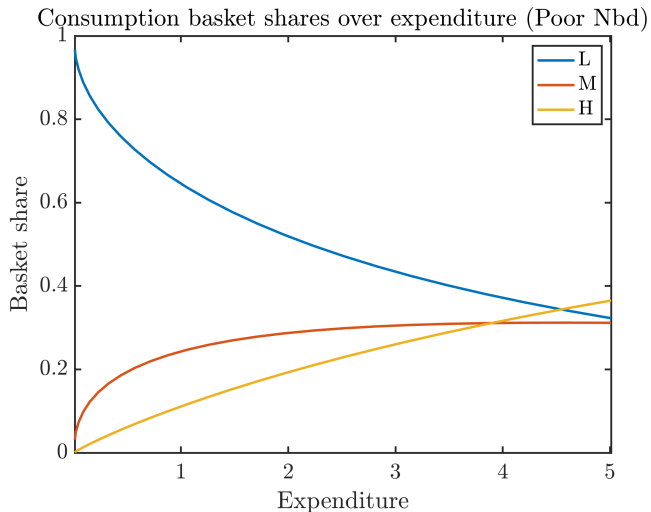
Targeted moments

Moment	Data	Model	Source
ROI income	10.0%	8.3%	Boar and Giannone (2025)
ROI grocery	10.0%	12.1%	
Rent/Wage Periphery	0.34	0.36	Own calculations, (Nielsen, CBP, ACS)
Rent/Wage Core	0.33	0.47	
Core/Periphery population	1.04	1.23	
Core/Periphery housing stock	0.96	0.91	
Core/Periphery store density	2.23	1.28	
Elasticity of stores visited to store density	0.041	0.040	
Wealth/Income	3.00	2.43	Nord (2022)
Basket overlap (Q1 Q5)	63%	71%	
Average markup	1.39	1.40	
$\text{mean}(CoV_j)$	0.19	0.19	
$CoV_2 - CoV_1$	-0.012	-0.011	
$CoV_3 - CoV_1$	-0.020	-0.021	
Across variety savings (Q5 Q1)	7.23%	6.87%	
Within variety savings (Q5 Q1)	1.99%	1.92%	

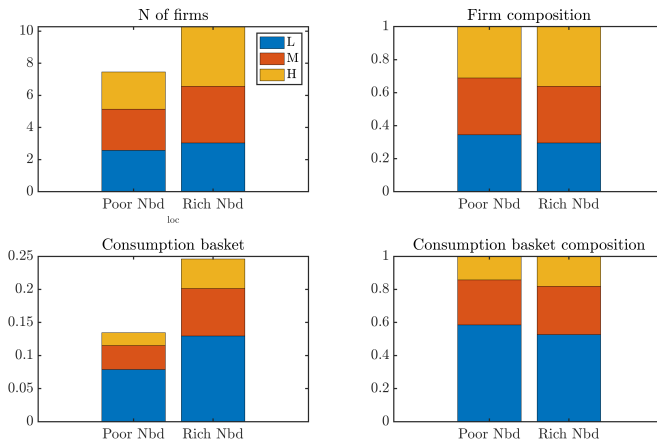
Calibrated model - Segregation



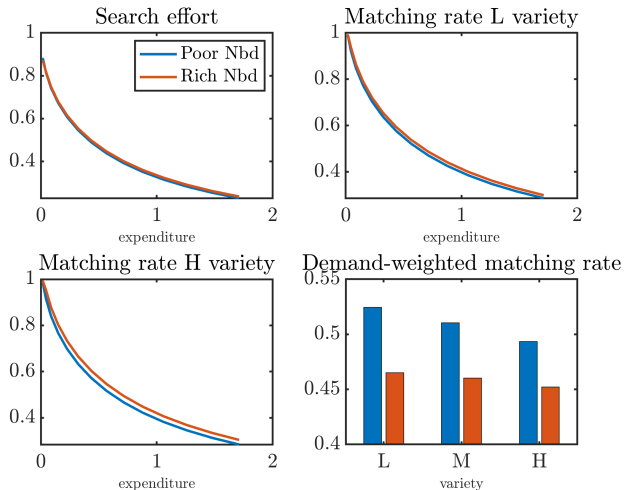
Calibrated model - Variety allocations



Calibrated model - Consumption sorting



Calibrated model - Search



Median markup in Rich Nbd is

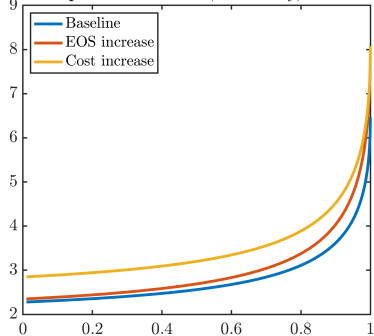
- 5.2% higher for variety L,
- 4.3% higher for variety M
- 3.4% higher for variety H.

p_G is 2.5% higher.

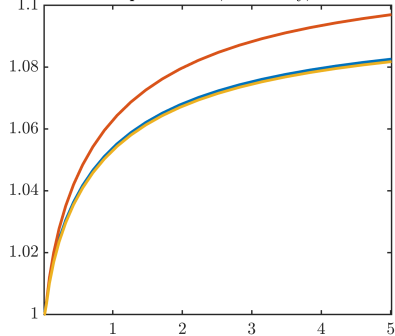
Calibrated model - Comparative Statics

- ① The monopolist price increases 25% through the EOS σ_j
- ② The monopolist price increases 25% through the marginal cost κ_j

Posted price distribution, H variety, Rich Nbd



Normalize price index, H variety, Rich Nbd



Counterfactual

Marginal costs increase uniformly by 25% in GE. Steady state comparison:

⇒ Price indexes increase 6 to 8%, rent decreases by 2%.

⇒ Compute increase in *cost of living*, expenditure needed to maintain same flow utility:

	Poor Nbd	Rich Nbd	
Bottom 50%	4.3%	3.7%	4.0%
50-75	4.6%	3.9%	4.2%
75-90	4.7%	4.0%	4.3%
Top 10%	4.7%	4.0%	4.3%
	4.4%	3.9%	

Table 1: Increase in cost of living, across wealth distribution and locations.

Conclusions

- Developed a spatial model of a frictional product market with heterogeneous hhs;
- Model allows to study different shopping behaviors and their externalities;
- Provided empirical evidence of the model's main mechanisms;
- An increase in marginal costs has non-trivial, different impacts across spatial and wealth distributions.

⇒ Looking beyond income segregation into consumption sorting uncovers role of shopping externalities, aggregation, and congestion forces.

$$\bar{C}_j = \int \lambda(e) [p_j(s(e))]^{\sigma_j} c_j(e) de \quad (1)$$

$$\bar{m}_j = \int \frac{\lambda(e) [p_j(s(e))]^{\sigma_j} c_j(e)}{\bar{C}_j} m(s(e)) de \quad (2)$$

Search intensity on store density (per dollar)

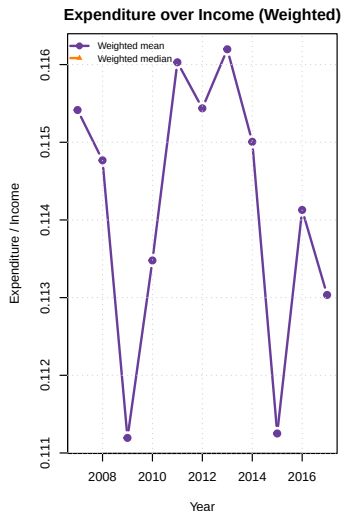
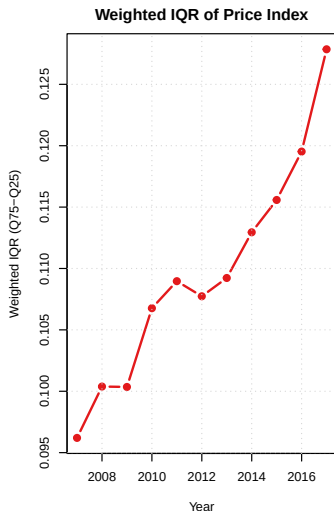
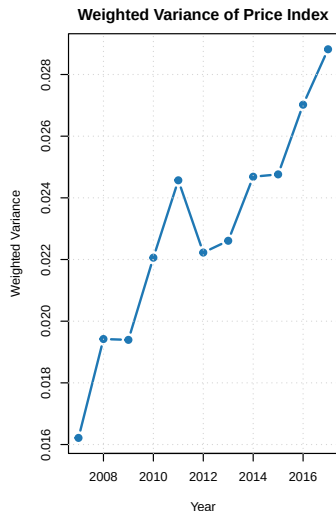
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:	log_stores_per_dollar		log_trips_per_dollar	
<i>Variables</i>				
log_nord_stores_per_sqkm	0.0465*** (0.0017)	0.0450*** (0.0019)	0.0201*** (0.0016)	0.0214*** (0.0019)
log_income	-0.1564*** (0.0050)	-0.1643*** (0.0051)	-0.1888*** (0.0048)	-0.1829*** (0.0048)
ln_county_income_hh		0.0653*** (0.0135)		-0.0864*** (0.0129)

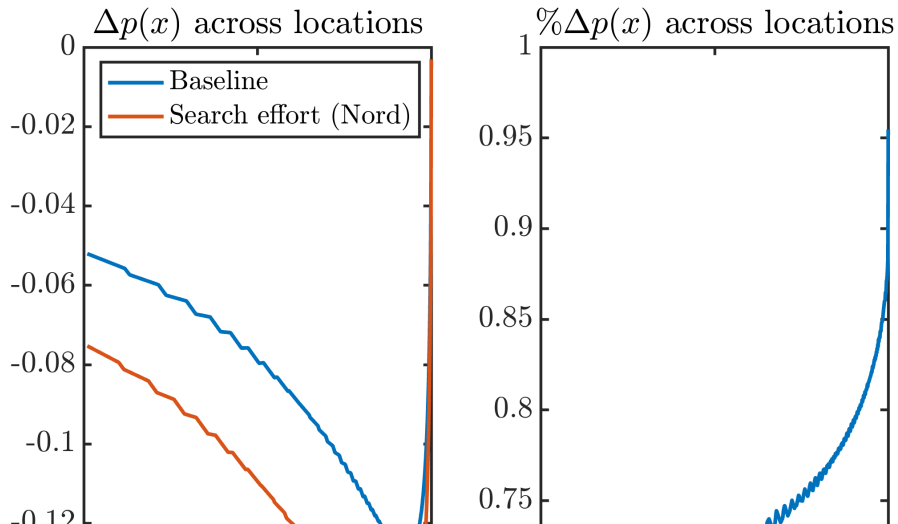
Prices on search intensity [Back](#)

	Log(Price Index)			
Log(Stores Visited)	-0.0257*** (0.0006)	-0.0259*** (0.0006)		
Log(Trips)			-0.0343*** (0.0006)	-0.0343*** (0.0006)
Log(Income)	0.0168*** (0.0008)	0.0162*** (0.0008)	0.0124*** (0.0008)	0.0120*** (0.0008)
Log(Expenditure)	0.0292*** (0.0009)	0.0293*** (0.0009)	0.0472*** (0.0009)	0.0471*** (0.0009)
Log(County Income)		0.0122*** (0.0025)		0.0072*** (0.0024)
Observations	6,879,991	6,879,991	6,879,991	6,879,991
R ²	0.05355	0.05380	0.07000	0.07009

Price dispersion over time



Nord exercise



References I

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